

Problem

A dc voltmeter with a sensitivity of $10 \text{ k}\Omega/\text{V}$ is used to find the Thevenin equivalent of a linear network. Readings on two scales are as follows:

(a) 0–10 V scale: 8 V

(b) 0–50 V scale: 10 V

Obtain the Thevenin voltage and the Thevenin resistance of the network.

Step-by-step solution

Step 1 of 7

The sensitivity of a dc voltmeter is $10 \text{ k}\Omega/\text{V}$. Then current I_{FS} is,

$$\begin{aligned} I_{\text{FS}} &= \frac{1}{\text{sensitivity}} \\ &= \frac{1}{\left(\frac{10 \text{ k}\Omega}{\text{volt}} \right)} \\ &= 100 \mu\text{A} \end{aligned}$$

Step 2 of 7

Calculate the resistance R_M for the 0–10 V scale.

$$\begin{aligned} R_M &= \frac{V_{\text{FS}}}{I_{\text{FS}}} \\ &= \frac{10}{100} \\ &= 100 \text{ k}\Omega \end{aligned}$$

Calculate the resistance R_M for the 0–50 V scale.

$$\begin{aligned} R_M &= \frac{V_{\text{FS}}}{I_{\text{FS}}} \\ &= \frac{50}{100} \\ &= 500 \text{ k}\Omega \end{aligned}$$

Step 3 of 7

Thevenin equivalent of the linear network is,

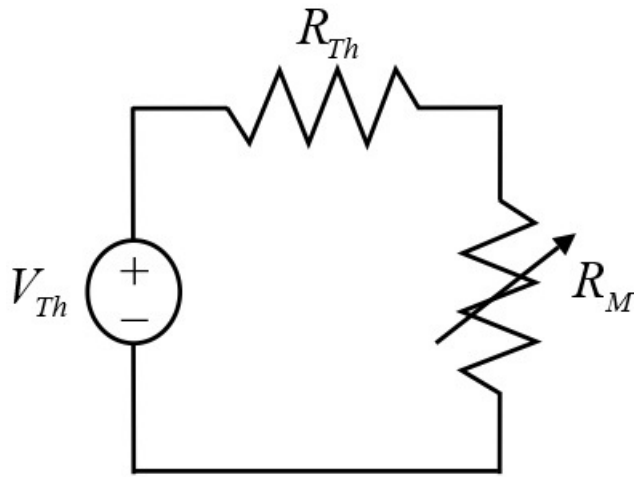


Figure 1

Step 4 of 7

The Thevenin's voltage in the Figure 1 is,

$$V_{Th} = I(R_M + R_{Th})$$

Calculate the current I for the 0–10 V scale.

$$\begin{aligned} I &= \frac{V}{R} \\ &= \frac{8}{100 \text{ k}} \\ &= 80 \mu\text{A} \end{aligned}$$

The Thevenin's voltage is,

$$V_{Th} = 80 \mu\text{A} \times R_{Th} + 8 \dots\dots (1)$$

Step 5 of 7

Calculate the current I for the 0–50 V scale.

$$\begin{aligned} I &= \frac{V}{R} \\ &= \frac{10}{500 \text{ k}} \\ &= 20 \mu\text{A} \end{aligned}$$

The Thevenin's voltage is,

$$V_{Th} = 20 \mu\text{A} \times R_{Th} + 10 \dots\dots (2)$$

Step 6 of 7

From equation (1) and (2).

$$80 \times 10^{-6} \times R_{Th} + 8 = 20 \times 10^{-6} \times R_{Th} + 10$$

$$60 \times 10^{-6} \times R_{Th} = 2$$

$$R_{Th} = 33.33 \text{ k}\Omega$$

Hence, the Thevenin resistance is $33.33 \text{ k}\Omega$.

Step 7 of 7

Substitute $33.3 \text{ k}\Omega$ for R_{Th} in equation (2).

$$V_{Th} = (20 \times 10^{-6})(33.33 \times 10^3) + 10$$

$$= 666.6 \times 10^{-3} + 10$$

$$= 10.667 \text{ V}$$

Thus, the Thevenin voltage is 10.667 V .